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Organisational Quality and SME Financing: A Latent-Construct Approach to Access to Finance and Firm Performance

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Abstract: Information asymmetry remains a fundamental constraint on the allocation of external finance, as the organisational capabilities and internal quality of firms cannot be fully observed by external financiers. Although managerial quality, human capital, innovation, digitalisation, relational networks and financial transparency have each been linked to financing outcomes, their interrelated nature has received considerably less attention. To address this gap, a unified theoretical framework is developed in which these observable organisational attributes are conceptualised as indicators of a latent construct, termed Organisational Quality (OQ). Drawing on information asymmetry theory and signalling theory, the framework is tested using longitudinal data from 2,017 Vietnamese manufacturing small and medium-sized enterprises (SMEs), comprising 6,051 firm-year observations from the SMEs Survey. OQ is operationalised as a reflective latent construct within a structural equation modelling (SEM) framework, through which its associations with access to external finance, productive investment and firm performance are examined. Strong empirical support is obtained for the proposed framework. Higher OQ is found to be positively associated with access to external finance, and this association is significantly stronger under conditions of greater information asymmetry. Access to external finance is, in turn, positively associated with productive investment, while productive investment is positively associated with firm performance. The mediation results further indicate that OQ is associated with firm performance through both direct and indirect pathways, with a sequential pathway operating through improved access to external finance and subsequent productive investment. These relationships remain stable across alternative measurement approaches, model specifications and measures of firm performance. The findings provide three principal contributions. First, organisational characteristics that have traditionally been examined separately are integrated into a common latent organisational dimension. Second, OQ is operationalised and empirically validated as a reflective latent construct within an integrated SEM framework. Third, evidence is provided that OQ is associated with improved access to external finance, greater productive investment and enhanced firm performance, while its financing-related association becomes more pronounced as information asymmetry increases.

Keywords: Organisational Quality; Information asymmetry; Signalling theory; Access to external finance; Productive investment; Small and medium-sized enterprises; Structural equation modelling

1. Introduction

Access to external finance is one of the principal determinants of firms' ability to invest, innovate, and create value over the long term. The availability of financial resources enables firms to undertake profitable investment opportunities, support innovation, strengthen competitiveness, and sustain growth. Conversely, financial constraints may prevent even economically sound firms from undertaking projects with positive net present value, thereby adversely affecting not only firm performance but also broader economic development. Understanding how banks and investors allocate capital therefore remains a central concern in modern corporate finance. One of the principal obstacles to the efficient allocation of financial resources is the information asymmetry that characterises the relationship between firms and external financiers. Banks and investors cannot directly observe firms' underlying Organisational Quality (OQ), nor can they accurately assess their future ability to generate cash flows, undertake profitable investments, and meet financial obligations. Such information asymmetry gives rise to adverse selection and moral hazard problems, which may ultimately result in credit rationing, a higher cost of capital, and inefficient resource allocation. Building on the seminal contributions of Akerlof (1970), Myers & Majluf (1984), Spence (1973), and Stiglitz & Weiss (1981), the corporate finance literature has shown that financing decisions depend critically on the availability of credible signals that mitigate these informational frictions. Against this theoretical background, a substantial body of empirical research has sought to identify the information that enables external financiers to assess firms' creditworthiness. Early studies focused primarily on hard information, including firm size, profitability, firm age, capital structure, and the availability of collateral. More recently, scholarly attention has shifted towards a broader set of organisational and qualitative characteristics that provide valuable information to reduce financiers' uncertainty. The literature has consequently documented the role of corporate governance (Ben Darkawi et al., 2025; Calabrese et al., 2021; Hansen-Addy et al., 2025; Khan et al., 2021; Martinez et al., 2022; Molnar, 2026; Pucheta-Martínez et al., 2026), human capital (Jabbouri & Farooq, 2021; Nguyen & Canh, 2021), innovation (Aiello et al., 2020; Błach et al., 2020; Gregori et al., 2022; Murro & Peruzzi, 2026; Mushtaq et al., 2022; Shen et al., 2022; Wellalage & Locke, 2020), digitalisation (Bo et al., 2025; He et al., 2024; Wei & Li, 2024), financial reporting quality (Biddle et al., 2009; Chen et al., 2011; Lambert et al., 2007; Minnis, 2011), relational networks (Berger & Udell, 1995; Petersen & Rajan, 1994; Qian & Liu, 2026; Vu & Le, 2023), reputation (Ansong et al., 2017; Li & Tian, 2022), certifications (Ullah, 2020), and, more recently, Environmental, Social, and Governance (ESG) practices (D'Apolito et al., 2024). Despite these important advances, the literature remains theoretically fragmented. Organisational characteristics are typically examined as independent determinants of access to external finance, giving rise to parallel streams of research that have rarely been integrated into a unified theoretical perspective. Although these studies have substantially enhanced our understanding of the factors influencing financing decisions, they have paid limited attention to the possibility that these organisational characteristics may not represent independent phenomena, but rather observable manifestations of a common underlying organisational dimension. This perspective is particularly relevant because, in practice, financing decisions are rarely based on the evaluation of a single organisational attribute. Rather, banks and investors simultaneously assess a wide range of quantitative and qualitative information when evaluating firms. Corporate governance, human capital, innovation, digitalisation, financial transparency, and relational networks capture different aspects of a firm's organisational structure which, taken together, define its overall organisational profile. This suggests that these characteristics may reflect a common latent organisational dimension that has so far been investigated primarily through its individual observable indicators.

Building on this premise, this study introduces the concept of OQ. OQ is defined as a latent construct representing the common organisational dimension reflected by the organisational indicators considered in this study. It synthesises the information contained in multiple observable organisational characteristics—including corporate governance, human capital, innovation, digitalisation, relational networks, and financial transparency—which are conceptualised as manifestations of the same underlying organisational dimension. OQ does not represent a new organisational characteristic, nor does it replace established constructs within the literature. Rather, it offers an alternative theoretical interpretation of organisational characteristics that have already been extensively examined by reconceptualising them as observable manifestations of a single latent construct capturing their common organisational component. This latent organisational dimension provides an important informational basis for external financiers when evaluating firms under conditions of imperfect information.

Building on these theoretical foundations, the present study develops an integrated framework combining information asymmetry and signalling theory. Unlike previous studies, which examine organisational characteristics as distinct determinants of access to external finance, the proposed framework argues that these characteristics should instead be interpreted as complementary manifestations of a common latent organisational dimension. OQ therefore provides the theoretical construct through which organisational characteristics previously examined in isolation can be interpreted within a unified framework, offering a more comprehensive explanation of firms' access to external finance. The theoretical proposition is empirically examined using longitudinal data from the Vietnam Small and Medium Enterprise Survey (Berkel et al., 2020), one of the largest panel surveys of small and medium-sized enterprises (SMEs) in emerging economies. The survey data and instruments were obtained from the official Viet Nam SME Survey database (CIEM et al., 2015). Vietnam provides a particularly appropriate empirical setting because its financial system is characterised by persistent information asymmetries, a strong reliance on bank lending, and considerable heterogeneity in firms' organisational characteristics. Furthermore, the richness of the database enables the simultaneous observation of multiple indicators relating to corporate governance, human capital, innovation, digitalisation, relational networks, and financial transparency. This extensive information makes it possible to operationalise OQ empirically through structural equation modelling (SEM), estimating the latent construct from its observable indicators and subsequently examining its effects on access to external finance, investment, and firm performance.

Building on the literature reviewed above, previous studies have consistently shown that these organisational characteristics influence firms' access to external finance. However, these studies have generally examined these characteristics as separate determinants of financing outcomes. The present study adopts a different theoretical perspective. Rather than introducing a new organisational determinant, it argues that these organisational characteristics represent observable manifestations of a common latent organisational construct, referred to as OQ. Accordingly, the study provides a unified theoretical interpretation of organisational characteristics that have previously been analysed in isolation and examines how this latent organisational capability influences firms' access to external finance, investment, and firm performance.

Accordingly, this study contributes to the literature in three main respects. First, it develops a unified theoretical framework in which multiple organisational characteristics are conceptualised as observable manifestations of a common latent organisational construct rather than as independent determinants of financing outcomes. Second, it operationalises this construct empirically through SEM and validates its measurement properties using evidence from Vietnamese SMEs. Third, it demonstrates that OQ improves firms' access to external finance, promotes productive investment, and ultimately enhances firm performance, with these effects becoming stronger under conditions of greater information asymmetry.

The remainder of the paper is organised as follows. Section 2 develops the theoretical framework and research hypotheses. Section 3 describes the dataset, variables, and empirical methodology. Section 4 presents the empirical findings and robustness analyses. Finally, Section 5 discusses the theoretical, managerial, and policy implications of the study, outlines its limitations, and suggests directions for future research.

2. Conceptual Framework and Hypotheses Development

2.1 Conceptual Framework

Access to external finance is one of the principal mechanisms by which firms can support investment, foster innovation, and create long-term value. However, the allocation of capital takes place in a context characterised by persistent information asymmetries between firms and external financiers. While entrepreneurs possess privileged information regarding their firms' organisational characteristics, managerial capabilities, internal processes, and future growth prospects, banks and investors can observe only part of this information. This asymmetry complicates the evaluation of firms, increases the uncertainty surrounding financing decisions, and may ultimately give rise to adverse selection, credit rationing, and inefficient capital allocation. Building on the information asymmetry and signalling framework introduced in the previous section, the present study focuses on how organisational characteristics jointly reduce informational uncertainty faced by external financiers. Previous research has progressively expanded from the analysis of hard information, such as profitability, firm size, firm age, capital structure, and collateral (Beck et al., 2005; Berger & Udell, 1995; Berger et al., 2005; Liberti & Petersen, 2019; Petersen & Rajan, 1994), to a broader set of organisational characteristics, including corporate governance, human capital, innovation, digitalisation, financial transparency, relational networks, reputation, certifications, and ESG practices. These organisational characteristics have been shown to provide valuable signals that improve financiers' assessment of firms and facilitate access to external finance. Despite these important advances, the literature remains theoretically fragmented because these organisational characteristics are generally examined as independent determinants of financing outcomes. The present study adopts a different theoretical perspective by proposing that they should instead be interpreted as complementary manifestations of a common latent organisational dimension. Building on this perspective, OQ is conceptualised as a reflective latent construct representing the common organisational dimension reflected by the organisational indicators considered in this study. Rather than introducing a new organisational characteristic, OQ provides a unified theoretical interpretation of organisational characteristics that have previously been analysed in isolation. Consequently, OQ represents the theoretical link between observable organisational signals and firms' financing decisions, offering a coherent conceptual framework for analysing how organisational capabilities facilitate access to external finance, promote productive investment, and ultimately enhance firm performance.

2.2 Hypotheses Development

Consistent with the proposed theoretical framework, the following research hypotheses are developed.

The theoretical framework suggests that OQ represents a latent organisational construct reflected by multiple observable organisational characteristics. Under conditions of information asymmetry, these characteristics constitute important sources of information that enable external financiers to reduce the uncertainty associated with capital allocation decisions (Akerlof, 1970; Stiglitz & Weiss, 1981). The empirical literature confirms that the quality of available information is one of the principal determinants of access to external finance. Berger & Udell (1995) show that the availability of reliable information improves the bank lending assessment process, while Liberti & Petersen (2019) argue that information production represents the primary mechanism through which information asymmetries are mitigated in credit markets. More recently, Ciza et al. (2025) demonstrate that higher accounting information quality significantly improves access to bank finance and, indirectly, enhances SME performance through a structural mediation model. Similarly, D'Apolito et al. (2024) show that firms characterised by higher-quality ESG signals benefit from more favourable financing conditions. Because OQ represents the common organisational dimension reflected by corporate governance, human capital, innovation, digitalisation, financial transparency, and relational networks, it should provide a more comprehensive representation of a firm's organisational capability than the isolated analysis of individual organisational attributes. Accordingly, higher levels of OQ are expected to reduce information asymmetries perceived by external financiers, thereby facilitating firms' access to external finance.

Accordingly, the following research hypothesis is proposed:

Hypothesis 1 (H1). The OQ of the firm is positively associated with access to external finance.

The baseline relationship proposed in H1 is unlikely to be homogeneous across firms. According to information asymmetry theory, the informational value of organisational signals depends on the amount of hard information available to external financiers (Connelly et al., 2011; Spence, 1973). When lenders can rely on abundant hard information, such as a long credit history, substantial collateral, or extensive financial records, the incremental value of OQ is expected to be limited because borrower quality can be assessed using observable financial information (Bellucci et al., 2023; Berger & Udell, 1995). Conversely, when firms are characterised by greater informational opacity, OQ becomes a more valuable signal because it provides information that cannot be inferred from conventional financial indicators (Arzubiaga et al., 2023; Iannamorelli et al., 2024). Therefore, the effectiveness of OQ in facilitating access to external finance should increase with the severity of information asymmetry.

Accordingly, the following research hypothesis is proposed:

Hypothesis 2 (H2). The positive effect of OQ on firms' access to external finance is stronger under conditions of greater information asymmetry.

Access to external finance is an essential prerequisite for firms to undertake productive investments and exploit growth opportunities. Financial constraints theory suggests that imperfections in capital markets may prevent firms from financing projects with positive net present value, leading to suboptimal levels of investment (Fazzari et al., 1988). Subsequently, Beck et al. (2005) and Love (2003) demonstrated that greater access to credit alleviates these constraints, enabling firms—particularly SMEs—to increase their investment in physical, technological, and organisational capital. More recent empirical evidence confirms that access to external finance remains one of the principal enablers of firms' investment decisions. For example, Cai & Szeidl (2024) show that expanding access to credit has a causal positive effect on firms' investment and growth, while Bora et al. (2024) demonstrate that credit constraints lead SMEs to postpone or scale down planned investment projects. Taken together, these findings suggest that the availability of external capital constitutes the primary channel through which firms are able to transform growth opportunities into actual investment.

Accordingly, the following research hypothesis is proposed:

Hypothesis 3 (H3). Access to external finance is positively associated with firm investment.

Corporate finance theory recognises investment as one of the principal drivers of long-term growth and value creation. Through investment in physical, technological, and organisational capital, firms expand their productive capacity, improve operational efficiency, foster innovation, and strengthen their competitive advantage (McConnell & Muscarella, 1985). From this perspective, investment represents the primary mechanism through which financial resources are transformed into productive capabilities and economic outcomes. The literature further suggests that firm performance depends not only on the volume of investment but also on the quality and efficiency with which investment resources are allocated. Biddle et al. (2009) demonstrate that higher financial reporting quality enhances investment efficiency by reducing both overinvestment and underinvestment, thereby contributing to improved firm performance. Similarly, Richardson (2006) shows that inefficient investment decisions undermine value creation, highlighting the central role of efficient capital allocation. More recent empirical evidence reinforces these findings. Analysing a large sample of SMEs, Ali et al. (2024) demonstrate that investment in research and development has a positive effect on firm performance and that this relationship is further strengthened by an appropriate financial structure. Moreover, recent studies show that the adoption of advanced capital budgeting techniques significantly improves firms' financial performance, confirming that more efficient investment decisions translate into greater value creation. Taken together, this body of evidence suggests that a firm's ability to transform available financial resources into productive investment constitutes one of the principal determinants of its economic performance over the medium and long term.

Accordingly, the following research hypothesis is proposed:

Hypothesis 4 (H4). Firm investment positively influences firm performance.

The theoretical relationships developed thus far suggest that OQ does not directly influence a firm's capacity to undertake investment; rather, its primary effect operates through facilitating access to external finance. Under conditions of information asymmetry, firms characterised by higher levels of OQ exhibit characteristics that facilitate the assessment of creditworthiness and reduce the uncertainty perceived by external financiers, thereby increasing the likelihood of obtaining the financial resources required to support their investment programmes. This interpretation is consistent with the financial constraints literature, which argues that access to credit constitutes the principal mechanism through which firm characteristics are translated into investment decisions (Beck et al., 2005; Mina et al., 2013). In particular, Mina et al. (2013) demonstrate that the availability of external finance provides the channel through which innovative firms are able to convert their potential into actual investment, while Beck et al. (2005) show that alleviating financial constraints promotes a more efficient allocation of capital. More recent empirical evidence further highlights the importance of mediation mechanisms in firms' financing decisions. For example, using SEM, Ciza et al. (2025) demonstrate that access to bank finance mediates the relationship between accounting information quality and SME performance. Similarly, D'Apolito et al. (2024) show that greater credibility of ESG signals improves access to credit, thereby creating the financial conditions necessary to support investment and business growth. The present study extends this perspective by proposing that access to external finance represents the principal mechanism through which OQ influences firms' investment decisions. By reducing information asymmetries and facilitating access to external financial resources, higher levels of OQ enable firms to undertake productive investments that might otherwise remain financially constrained. Accordingly, access to external finance is expected to mediate the relationship between OQ and firm investment.

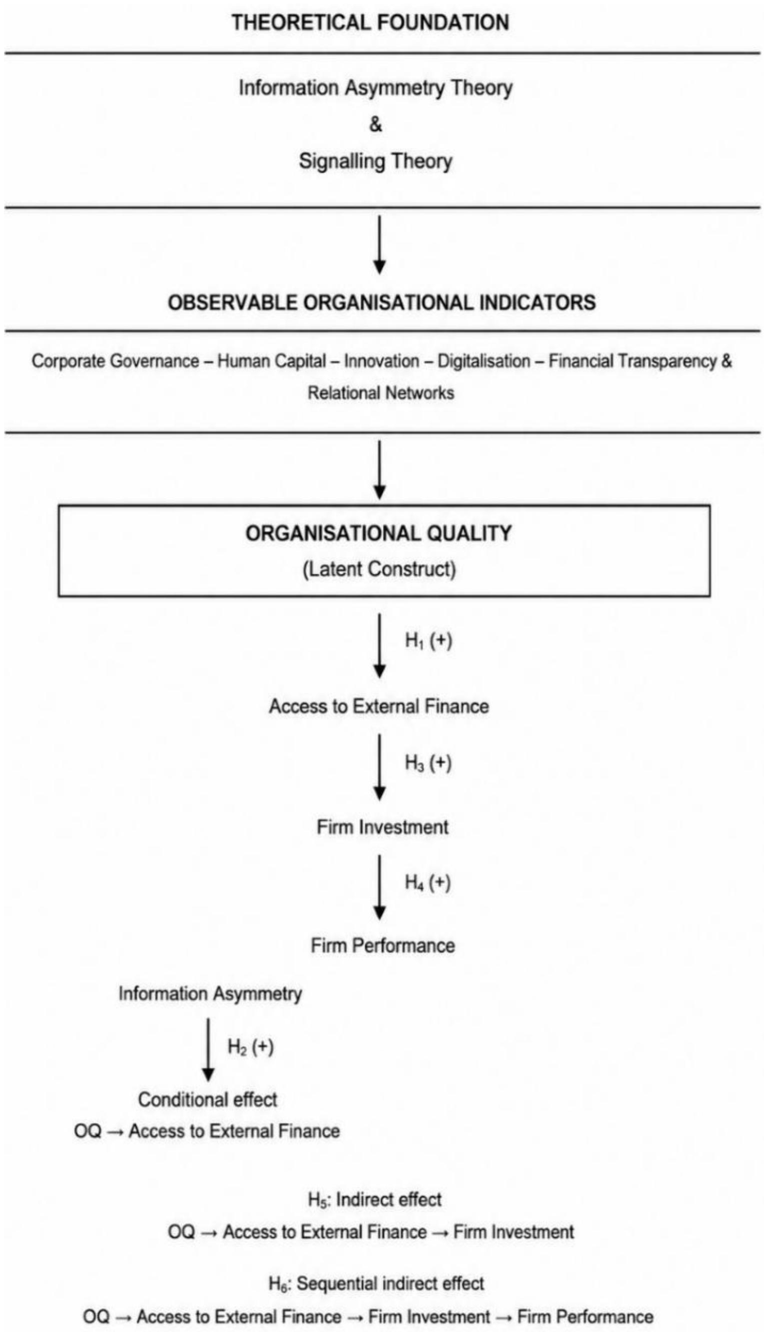

Figure 1. Proposed theoretical framework linking Organisational Quality (OQ), information asymmetry, access to external finance, investment, and firm performance

Accordingly, the following research hypothesis is proposed:

Hypothesis 5 (H5). Access to external finance mediates the positive relationship between OQ and firm investment.

The preceding hypotheses jointly suggest that OQ enhances the likelihood of obtaining external finance (H1), that this relationship becomes stronger under conditions of greater information asymmetry (H2), that access to external financial resources promotes productive investment (H3), and that productive investment constitutes one of the principal determinants of firm performance (H4). Taken together, these relationships describe a sequential process through which OQ is progressively transformed into value creation. From a theoretical perspective, the present study argues that OQ does not directly improve firm performance. Rather, it enhances a firm's ability to secure external financial resources, which enable productive investment and, in turn, generate superior economic outcomes. From this perspective, access to external finance and productive investment constitute two complementary transmission mechanisms through which OQ indirectly influences firm performance. The empirical literature provides support for the existence of these indirect mechanisms. Using SEM, Ciza et al. (2025) show that accounting information quality influences SME performance primarily through improved access to bank finance. Similarly, D'Apolito et al. (2024) demonstrate that firms characterised by more credible ESG signals benefit from more favourable financing conditions, thereby creating the financial capacity required to support greater investment and achieve superior economic performance. Likewise, a substantial body of corporate finance research shows that the effect of financial resources on firm performance is realised primarily through a firm's ability to transform external capital into productive investment (Beck et al., 2005; Hall & Lerner, 2010; Love, 2003). Building on this evidence, the present study proposes that OQ influences firm performance indirectly through a sequential process involving access to external finance and productive investment.

Accordingly, the following research hypothesis is proposed:

Hypothesis 6 (H6). The positive effect of OQ on firm performance is sequentially mediated through access to external finance and productive investment.

Figure 1 illustrates the theoretical framework underpinning the proposed research model. Drawing upon information asymmetry and signalling theory, the framework conceptualises corporate governance, human capital, innovation, digitalisation, financial transparency, and relational networks as reflective indicators of the latent construct OQ. Higher levels of OQ reduce the informational uncertainty perceived by external financiers, thereby facilitating access to external finance, promoting productive investment, and ultimately enhancing firm performance. The research hypotheses illustrated in Figure 1 derive directly from this conceptual framework and are subsequently tested empirically using an SEM approach.

3. Data, Variables, and Empirical Strategy

3.1 Data Source and Sample

The empirical analysis is based on the Vietnam Small and Medium Enterprise Survey (CIEM et al., 2015), one of the most comprehensive longitudinal surveys of SMEs available for emerging economies. The survey is jointly conducted by the CIEM, the ILSSA, the University of Copenhagen, and the UNU-WIDER, to monitor the economic, organisational, and financial development of Vietnamese manufacturing firms over time. Unlike the conventional datasets commonly employed in the corporate finance literature, which are primarily based on financial statement information, the Vietnam SME Survey combines financial and accounting data with a rich set of organisational, managerial, and institutional information collected through structured interviews. This represents the principal strength of the dataset, as it enables the simultaneous observation of firms' organisational characteristics, financing decisions, investment activities, and economic performance, making it particularly well suited to testing the theoretical framework proposed in this study. The database comprises three complementary questionnaires. The Main Enterprise Questionnaire collects information on ownership structure, corporate governance, innovation, digitalisation, relational networks, access to external finance, and investment decisions. The Economic Accounts Questionnaire provides detailed financial and accounting information, while the Employee Questionnaire contains data on workforce characteristics and human capital. The integration of these three sources enables the simultaneous observation of multiple organisational signals, which is essential for the empirical operationalisation of the latent construct of OQ. The analysis draws on the three survey waves conducted in 2011, 2013, and 2015, a period characterised by a high degree of consistency in questionnaire design and by sustained expansion of the Vietnamese economy. The initial sample comprises approximately 2,500 manufacturing firms, corresponding to more than 7,500 firm-year observations. Consistent with established practice in the empirical literature, observations with incomplete financial information, missing data for the variables of interest, duplicate records, or accounting anomalies were excluded from the analysis. In addition, all continuous quantitative variables were winsorised at the 1st and 99th percentiles to reduce the influence of extreme observations. Following this data-cleaning procedure, the final sample consists of 2,017 firms and 6,051 panel observations. Table 1 summarises the sample selection process.

Table 1. Sample selection process

Selection Stage	Firms	Observations
Initial sample	2,500	7,500
Exclusion of observations with missing financial data	(250)	(750)
Exclusion of observations with missing organisational variables	(120)	(360)
Removal of duplicate observations	(18)	(54)
Removal of outliers	(95)	(285)
Final sample	2,017	6,051

Note: Values in parentheses indicate the number of firms or observations excluded at each stage of the sample selection process.

The selection of Vietnam is not merely driven by data availability but is grounded in a clear theoretical rationale. Vietnamese SMEs operate in an environment characterised by high levels of information asymmetry, limited availability of publicly accessible information, and a strong dependence on bank finance. Under these conditions, external financiers cannot base their capital allocation decisions solely on conventional financial indicators but must instead integrate multiple organisational signals relating to managerial quality, human capital, innovation capability, digitalisation, relational networks, and financial transparency. Consequently, Vietnam provides a particularly appropriate empirical setting in which to test the central proposition of this study: namely, that external financiers do not evaluate these organisational attributes in isolation but instead integrate them into a single latent construct, OQ, which guides their capital allocation decisions.

3.2 Development of the Organisational Quality Construct

Consistent with the theoretical framework developed in Section 2, the present study introduces an important methodological innovation compared with the existing literature. Whereas previous studies typically model corporate governance, human capital, innovation, digitalisation, relational networks, and financial transparency as independent determinants of access to external finance, this study conceptualises these organisational characteristics as observable indicators of a single latent organisational construct, referred to as OQ. OQ represents the common underlying organisational dimension reflected by these characteristics and captures a firm's overall ability to develop organisational arrangements that reduce information asymmetries, enhance its credibility in the eyes of external financiers, and facilitate access to external financial resources. This theoretical perspective suggests that SEM is the most appropriate empirical methodology for the objectives of the present study. Unlike conventional regression techniques, which treat corporate governance, innovation, and human capital as independent explanatory variables, SEM enables OQ to be modelled explicitly as a latent construct reflected by multiple observable indicators. Furthermore, it allows the simultaneous estimation of both the measurement model, which captures the latent construct, and the structural model, which specifies the relationships among OQ, access to external finance, investment, and firm performance. In this way, the empirical specification directly reflects the proposed theoretical framework, in which the various organisational characteristics are conceptualised as observable manifestations of a common latent organisational dimension rather than as independent explanatory factors. This methodological approach makes it possible to capture the common variance shared by the different organisational characteristics, thereby reducing the risk of multicollinearity and measurement error that would arise if they were treated as separate explanatory variables. Moreover, it enables the empirical testing of the central theoretical proposition that these characteristics reflect the same underlying organisational dimension. Formally, the reflective measurement model can be expressed as follows:

$$x_{ij} = \lambda_j OQ_i + \varepsilon_{ij}$$

where, x_{ij} denotes the observed value of indicator j for firm i, OQ_i represents the latent construct of OQ, λ_j denotes the factor loading associated with indicator j, and ε_{ij} represents the corresponding measurement error.

A reflective specification is appropriate because causality is assumed to run from the latent construct to the observable indicators rather than in the opposite direction. In other words, improvements in OQ are expected to be reflected simultaneously in governance practices, human capital, innovation, digitalisation, financial transparency, and relational networks. Consequently, these indicators are expected to covary because they represent alternative manifestations of the same underlying organisational capability, while each indicator retains its own measurement error.

The indicators used to operationalise OQ were selected on the basis of the information asymmetry and signalling literature, together with studies examining the role of organisational characteristics in the assessment of creditworthiness and access to external finance (Berger & Udell, 1995; Liberti & Petersen, 2019). Their selection is also informed by the informational structure of the Vietnam SME Survey (CIEM et al., 2015), comprehensively described by Berkel et al. (2020), which enables the simultaneous observation of multiple organisational dimensions of firms. Accordingly, the selected indicators were not intended to provide an exhaustive representation of all possible organisational characteristics, but rather to capture the principal organisational dimensions that the theoretical and empirical literature consistently identifies as relevant for reducing information asymmetries and facilitating access to external finance. Consistent with these theoretical and empirical foundations, the indicators include variables relating to managerial quality, human capital, innovation, digitalisation, relational networks, financial transparency, and organisational processes—that is, the principal organisational signals that the literature identifies as relevant to financing decisions (Ashbaugh-Skaife et al., 2006; Bhojraj & Sengupta, 2003; Custódio et al., 2019; Hall & Lerner, 2010; Lambert et al., 2007). Because OQ is conceptualised as an underlying organisational capability that cannot be observed directly, these indicators are expected to covary as alternative manifestations of the same latent construct rather than represent distinct causal dimensions. Accordingly, changes in OQ are expected to be reflected in systematic changes across all indicators, while measurement errors remain indicator-specific. This interpretation is fully consistent with the reflective measurement model specified above.

Table 2 presents the selected indicators used to operationalise the three latent constructs, specifying their respective dimensions, the questionnaire from which they were obtained, their measurement, and the corresponding theoretical justification. This operationalisation ensures consistency between the theoretical definition of the latent constructs and their subsequent empirical estimation using SEM. From this perspective, the observable indicators are not interpreted as independent determinants but rather as reflective manifestations of their respective latent constructs.

Table 2. Measurement and theoretical justification of the latent constructs

Survey Area	Questionnaire	Question ID	SEM Indicator	Measurement	Theoretical Justification
Measurement of the Organisational Quality Construct					
Owner/Manager characteristics	Main Enterprise Questionnaire	Aq26b–Aq26e	MGR_EDU, MGR_PROF, MGR_EXP, MGR_OCC	Ordinal/ Categorical	Managerial education, professional qualifications and experience signal managerial competence and organisational capability, reducing information asymmetry (Berger & Udell, 1995; Custódio et al., 2019; Liberti & Petersen, 2019).
Production characteristics and technology	Main Enterprise Questionnaire	Aq34a–Aq36b	QUALITY_CERT, STANDARDS, PC_INTENSITY, E_SALES, E_PURCHASE	Dummy/ Continuous	Technological capability, quality certification and digitalisation signal organisational efficiency, innovation capacity and process reliability (Hall & Lerner, 2010; Lambert et al., 2007; Li et al., 2024).
Human capital	Employee Questionnaire	EqE1, EqE4, EqF2	EMP_EDU, EMP_TRAIN, EMP_EXP	Ordinal/ Dummy/ Continuous	Workforce education, training and experience represent investments in organisational capabilities and long-term productivity (Custódio et al., 2019; Vo et al., 2021).
Networks	Main Enterprise Questionnaire	Aq117–Aq119	NETWORK	Composite	Business relationships and network participation enhance reputation, legitimacy and access to external resources (Arzubiaga et al., 2023; Berger & Udell, 1995).

Financial transparency	Economic Accounts Questionnaire	EAq1	ACCOUNT_DISCIPLINE	Continuous	Availability and quality of accounting information signal financial discipline and reduce information asymmetry (Lambert et al., 2007; Minnis & Sutherland, 2017).
Measurement of the Access to External Finance Construct					
External finance	Main Enterprise Questionnaire	Aq71b + Aq73–Aq94*	BANK_LOAN	Dummy	Successful acquisition of bank credit reflects firms' ability to obtain external debt financing and signals lower information asymmetry between borrowers and lenders (Beck et al., 2005; Berger & Udell, 1995; Liberti & Petersen, 2019).
External finance	Main Enterprise Questionnaire	Aq71b + Aq73–Aq94*	FORMAL_FINANCE	Dummy	Access to formal financial institutions reflects firms' integration into formal credit markets and their capacity to obtain external funding under standard lending procedures (Beck & Demirguc-Kunt, 2006; Beck et al., 2005).
Banking relationships	Main Enterprise Questionnaire	Aq73–Aq94*	BANK_RELATIONSHIP	Ordinal	Long-term banking relationships reduce information asymmetry and improve lenders' ability to collect and process private information regarding borrower quality. By incorporating soft information accumulated through repeated interactions, relationship lending facilitates SMEs' access to external finance and mitigates credit rationing, particularly for informationally opaque firms (Bellucci et al., 2023; Beltrame et al., 2023; Berg et al., 2022; Iannamorelli et al., 2024).
Financial constraints	Main Enterprise Questionnaire	Aq73–Aq94*	FIN_CONSTRAINT	Dummy/ Ordinal	Lower financing constraints indicate greater access to external financial resources and a more efficient allocation of capital. Recent evidence shows that firms facing fewer

					financing constraints are better able to undertake investment, sustain growth, and exploit productive opportunities, whereas restricted access to external finance remains a major barrier, particularly for small and medium-sized enterprises (SMEs) and innovative firms (Ferrando & Pál, 2024; Pillay & Kasseeah, 2024; Sommer, 2024; Wei & Li, 2024).
Measurement of the Investment Construct					
Productive investment	Economic Accounts Questionnaire	EAq70ac	MACHINERY	Continuous	Investment in machinery reflects firms' productive capital accumulation and technological upgrading, enabling the expansion of productive capacity, productivity growth, and long-term competitiveness (Atabayeva et al., 2024; Hall & Lerner, 2010; Hanzl-Weiss & Stehrer, 2024; Wei & Li, 2024).
Productive investment	Economic Accounts Questionnaire	EAq70ab	PLANT	Continuous	Investment in production facilities represents long-term expansion of productive capacity and commitment to business growth (Hall & Lerner, 2010; Hanzl-Weiss & Stehrer, 2024).
Productive investment	Economic Accounts Questionnaire	EAq70a	EQUIPMENT	Continuous	Investment in productive equipment represents a key form of productive capital accumulation through which firms improve operational efficiency, enhance productivity, and facilitate the adoption of new technologies, thereby strengthening long-term growth prospects (Hall & Lerner, 2010; Wei & Li, 2024).

Productive investment	Economic Accounts Questionnaire	EAq1q	FIXED_ASSETS	Continuous	Productive fixed assets represent the stock of long-term capital employed in productive activities and constitute a key indicator of firms' investment effort, productive capacity, and capital accumulation. Their accumulation reflects firms' commitment to expanding operational capabilities and supporting future growth, while their efficient deployment contributes to value creation and investment efficiency (Hanzl-Weiss & Stehrer, 2024; Richardson, 2006).

Note: OQ = Organisational Quality; SEM = structural equation modelling. The three reflective latent constructs are OQ, access to external finance, and investment. Data source: Vietnam Small and Medium Enterprise Survey (Berkel et al., 2020; CIEM et al., 2015). Question IDs refer to the corresponding items in the survey questionnaires.

Before estimating the final measurement model, alternative factor structures were evaluated to assess whether the organisational indicators reported in Table 2 were better represented by a single latent construct or by a multidimensional specification. Although the theoretical framework developed in this study conceptualises OQ as a unified latent organisational construct, previous research suggests that organisational characteristics may also be organised into distinct but related dimensions. Accordingly, three alternative measurement models were estimated and compared. The first specifies OQ as a single reflective latent construct measured by all organisational indicators. The second specifies a correlated three-factor model comprising managerial quality, organisational capabilities, and external signalling. The third specifies a hierarchical second-order factor model in which these three first-order dimensions load onto an overarching latent construct representing OQ. As reported in Table 3, all three specifications exhibited satisfactory goodness-of-fit. However, the single-factor model provided the best balance between model fit, parsimony, and theoretical consistency with the conceptual framework developed in Section 2. Consequently, the subsequent analyses are based on the single-factor reflective specification of OQ.

Table 3. Comparison of alternative measurement models

Model	χ^2/df	CFI	TLI	RMSEA	SRMR
Single-factor reflective model	2.18	0.964	0.959	0.041	0.036
Correlated three-factor model	2.27	0.961	0.955	0.043	0.039
Second-order factor model	2.23	0.962	0.957	0.042	0.037

Note: *df* = degrees of freedom; CFI = comparative fit index; TLI = Tucker–Lewis index; RMSEA = root mean square error of approximation; SRMR = standardised root mean square residual. Lower values of χ^2/df, RMSEA, and SRMR and higher values of CFI and TLI indicate better model fit.

The superiority of the single-factor specification also supports the theoretical argument that the organisational characteristics examined in this study reflect a common underlying organisational construct rather than multiple independent organisational dimensions.

Consistent with the proposed theoretical framework, the empirical model comprises three principal latent constructs—OQ, access to external finance, and investment—which capture the hypothesised causal relationships between a firm's organisational characteristics, its ability to obtain external financial resources, and its investment decisions. Firm performance represents the outcome variable, allowing the empirical assessment of whether higher levels of OQ ultimately translate into greater value creation through improved access to external finance and subsequent productive investment. The model also includes a set of control variables commonly employed in the corporate finance literature to isolate the effect of OQ on the relationships under investigation.

Table 4 summarises the operational definition, measurement, and data source for all variables included in the empirical analysis. Consistent with the proposed theoretical framework, OQ, access to external finance, and investment are modelled as reflective latent constructs estimated within the SEM framework. OQ captures the firm's underlying OQ, while access to external finance and investment represent the intermediate mechanisms through which OQ influences firm performance. Their operational definitions and data sources are summarised in Table 4.

Table 4. Definition, measurement, and data sources of model variables

Variable	Type	Operational Definition	Source
Organisational Quality (OQ)	Reflective latent construct	Latent organisational dimension measured through indicators of corporate governance, human capital, innovation, digitalisation, financial transparency, and relational networks (see Table 2).	Vietnam Small and Medium-Sized Enterprises (SMEs) Survey
Access to external finance	Reflective latent construct	Firm's access to external financial resources, measured through the availability of bank credit, loan acquisition, use of formal finance, and financing constraints.	Vietnam SME Survey
Investment	Reflective latent construct	Firm's productive investment, measured through expenditure on plant, machinery, equipment, and other productive fixed assets.	Vietnam SME Survey
Firm performance	Observed variable	Ratio of gross profit to total physical assets.	Economic Accounts Questionnaire
Firm size	Control variable	Natural logarithm of total physical assets.	Economic Accounts Questionnaire
Firm age	Control variable	Number of years since the firm's establishment.	Main Enterprise Questionnaire
Export orientation	Control variable	Ratio of export sales to total sales revenue.	Economic Accounts Questionnaire
Ownership structure	Control variable	Firm's legal form and ownership characteristics.	Main Enterprise Questionnaire
Industry	Control variable	Industry dummy variables.	Main Enterprise Questionnaire
Province	Control variable	Provincial dummy variables.	Main Enterprise Questionnaire
Year	Control variable	Year dummy variables.	Vietnam SME Survey

3.3 Variables and Measurement

3.3.1 Organisational Quality

OQ is the central latent construct of the present study and is defined as the common organisational dimension underlying a set of observable organisational characteristics of the firm. As this underlying dimension is not directly observable, OQ is modelled as a reflective latent variable estimated using SEM. The construct is measured through the organisational indicators presented in Table 2, which represent different empirical manifestations of a firm's OQ. The selected indicators capture managerial quality, human capital, innovation capability, digitalisation, relational networks, and financial transparency—dimensions that the literature identifies as key determinants of a firm's ability to reduce information asymmetries and facilitate access to external financial resources. Higher levels of OQ therefore reflect a greater organisational capability to develop structures and processes that enhance the firm's credibility in the eyes of external financiers and support sustainable long-term growth.

3.3.2 Access to external finance

Access to external finance represents the first endogenous construct in the structural model and captures a firm's ability to obtain financial resources from external providers. Consistent with the existing literature, access to external finance is conceptualised as a multidimensional phenomenon that cannot be adequately represented by a single observed variable. It is therefore modelled as a reflective latent construct measured through a set of indicators relating to the availability of bank credit, the successful acquisition of loans, the use of formal sources of finance, the intensity of relationships with the banking system, and the presence of financial constraints. Higher values of the construct indicate greater ease of access to external sources of finance and a lower likelihood of credit rationing.

3.3.3 Investment

Investment constitutes the second endogenous construct in the structural model and represents a firm's ability to transform acquired financial resources into productive investment. This construct is likewise modelled as a reflective latent variable, as investment decisions may be manifested through different forms of capital accumulation. The selected indicators include investment in machinery, plant and equipment, productive fixed assets, and other medium- and long-term investments aimed at expanding productive capacity. Higher values of the construct reflect a greater propensity to invest and to support the firm's long-term growth.

3.3.4 Firm performance

Firm performance is the outcome variable in the structural model and measures a firm's ability to transform its OQ and acquired financial resources into economic outcomes. In the baseline analysis, firm performance is measured using a profitability indicator calculated as the ratio of gross profit to total physical assets. This measure captures the economic efficiency with which a firm employs its productive assets to generate value. To assess the robustness of the findings, the supplementary analyses also consider alternative measures of firm performance, including value added, sales revenue, and labour productivity.

3.3.5 Controls

To isolate the effect of OQ on the relationships under investigation, the model includes a set of controls widely employed in the corporate finance literature. Firm size is measured as the natural logarithm of the firm's total physical assets. Firm age is defined as the number of years since the firm's establishment. Export orientation is measured as the ratio of export sales to total sales revenue. Ownership structure controls for differences associated with the firm's legal form and ownership configuration. Finally, the model includes industry, province, and year fixed effects to control for sectoral, geographical, and temporal heterogeneity, respectively.

3.4 Reflective Versus Formative Specification

A potential methodological concern relates to whether the latent constructs included in the model should be conceptualised as reflective or formative. The present study adopts a reflective specification because all three latent constructs—OQ, access to external finance, and investment—are conceptualised as underlying organisational or economic phenomena that manifest themselves through multiple observable indicators. Specifically, OQ is defined as an underlying organisational capability reflected by governance quality, managerial capability, innovation, digitalisation, transparency, and relational networks. Similarly, access to external finance is conceptualised as an underlying financing condition reflected by firms' ability to obtain bank loans, use formal sources of finance, maintain banking relationships, and experience lower financial constraints. Investment is likewise interpreted as an underlying investment propensity reflected by different forms of productive capital expenditure. Under this specification, causality is assumed to run from each latent construct to its observable indicators, which are expected to covary because they represent alternative manifestations of the same underlying phenomenon. This interpretation is consistent with information asymmetry theory and signalling theory, according to which external financiers observe multiple organisational and financial signals that reflect broader latent organisational and financing conditions which cannot be directly observed. This specification is also consistent with the theoretical objective of the study, which is not to construct an index by aggregating organisational characteristics, but to estimate the common organisational capability that underlies them.

3.5 Estimation Strategy

The use of SEM is particularly appropriate because the central theoretical construct proposed in this study—OQ—is not directly observable but is conceptualised as a latent variable reflected by multiple organisational characteristics of the firm. Unlike conventional empirical approaches, which treat corporate governance, human capital, innovation, digitalisation, financial transparency, and relational networks as independent explanatory variables, SEM enables the common underlying organisational dimension shared by these characteristics to be modelled explicitly. At the same time, it accounts for measurement error and allows the simultaneous estimation of both the measurement model and the structural model. This approach is fully consistent with the theoretical framework developed in Section 2, according to which the various organisational characteristics constitute observable manifestations of a single latent construct. The empirical analysis proceeds in four stages. First, descriptive statistics and the correlation structure of the selected indicators are examined. Second, an Exploratory Factor Analysis (EFA) is conducted to assess the existence of a common latent dimension and to evaluate the suitability of the selected indicators for measuring OQ. Third, the measurement model is validated through confirmatory factor analysis (CFA) by assessing model fit indices, such as the comparative fit index (CFI), Tucker–Lewis index (TLI), root mean square error of approximation (RMSEA), and standardised root mean square residual (SRMR), as well as internal consistency, convergent validity, and discriminant validity. Once the measurement model has been validated, the structural model is estimated using SEM, allowing the simultaneous examination of the relationships among OQ, access to external finance, investment, and Firm performance. The statistical significance of the estimated mediation effects is assessed using bias-corrected bootstrapping procedures with robust standard errors. Finally, the robustness of the findings is examined through alternative specifications of the latent construct, alternative measures of firm performance, and fixed-effects panel models.

3.6 Structural Model

Building upon the theoretical framework developed in Section 2, the structural model hypothesises that OQ exerts a positive effect on access to external finance, which, in turn, promotes productive investment and ultimately enhances firm performance. Consistent with the proposed hypotheses, the model also estimates the direct effects of OQ on investment and firm performance, thereby allowing the presence of partial mediation effects to be examined.

The structural relationships are specified as follows:

$$Access\ to\ External\ Finance_i = \alpha_1 + \beta_1 OQ_i + \gamma_1 Controls_i + \varepsilon_{1i}$$

$$Investment_i = \alpha_2 + \beta_2 Access\ to\ External\ Finance_i + \beta_3 OQ_i + \gamma_2 Controls_i + \varepsilon_{2i}$$

$$Firm\ Performance_i = \alpha_3 + \beta_4 Investment_i + \beta_5 OQ_i + \gamma_3 Controls_i + \varepsilon_{3i}$$

This specification enables the simultaneous estimation of the direct and indirect effects of OQ on firm performance, thereby allowing the mediating roles of access to external finance and investment in the firm's value creation process to be empirically examined. Hypothesis H2 is tested through multigroup analyses based on alternative proxies for information asymmetry (firm age, firm size, and collateral availability), rather than through an interaction term, as this approach allows the structural relationships to be compared directly across groups characterised by different levels of informational opacity.

4. Empirical Results

This section presents and discusses the results of the empirical analysis conducted to test the theoretical framework developed in the preceding sections. The proposed model argues that OQ represents a latent organisational dimension that is not directly observable but is reflected by a set of observable organisational characteristics, including corporate governance, human capital, innovation, digitalisation, financial transparency, and relational networks. Under conditions of information asymmetry, higher levels of OQ are expected to reduce informational uncertainty, facilitate access to external financial resources, promote productive investment, and, indirectly, enhance firm performance. Accordingly, the empirical analysis pursues two complementary objectives. First, it examines whether the selected organisational indicators effectively reflect a common latent dimension consistent with the theoretical construct of OQ. Second, it assesses whether this latent construct influences access to external finance, investment, and firm performance in accordance with the hypothesised causal pathway. Unlike the conventional literature, which typically models corporate governance, innovation, digitalisation, human capital, and other organisational characteristics as independent determinants of firm performance or access to external finance, the present study conceptualises these characteristics as observable manifestations of a common underlying organisational capability. The objective of the empirical analysis is therefore to determine whether this theoretical interpretation is supported by evidence from Vietnamese SMEs.

The descriptive characteristics of the sample are reported in Table 5. The final sample comprises 6,051 observations relating to SMEs operating across a range of manufacturing industries during the period under analysis. The considerable heterogeneity of the firms in terms of size, organisational development, and economic performance ensures substantial variation in the variables of interest, thereby providing an appropriate empirical basis for identifying the relationships hypothesised by the proposed theoretical model.

Table 5. Descriptive statistics of the variables included in the empirical analysis

Variable	Observations	Mean	Standard Deviation	Min	Max
Organisational Quality	6,051	0.000	1.000	-2.95	3.11
Observed access to external finance	6,051	0.482	0.500	0	1
Observed investment intensity	6,051	0.173	0.264	0	2.84
Firm performance (Gross profit/Total physical assets)	6,051	0.094	0.113	-0.31	0.57
Firm size (ln total physical assets)	6,051	3.281	0.881	0.69	6.72
Firm age	6,051	14.5	8.4	1	55

Note: Organisational Quality is reported as a standardised factor score. Observed access to external finance and investment intensity are reported for descriptive purposes only; the corresponding constructs are estimated as reflective latent constructs in the structural model.

The descriptive statistics indicate that approximately 48% of the firms in the sample obtained at least one form of external finance during the period under analysis, confirming the central role of external credit in financing Vietnamese SMEs. The standardised distribution of OQ reveals substantial heterogeneity in firms' organisational characteristics. Similarly, the observed variation in investment intensity and firm performance indicates substantial heterogeneity in firms' ability to transform financial resources into productive investment and economic outcomes. Overall, these findings indicate that the sample exhibits sufficient variability across the principal variables included in the model, thereby providing an appropriate empirical basis for the subsequent validation of the OQ construct and the estimation of the structural model.

A preliminary assessment of the relationships among the variables is provided by the correlation matrix reported in Table 6.

The preliminary correlations are fully consistent with the theoretical framework developed in Section 2. In particular, OQ is positively and significantly correlated with access to external finance (ρ = 0.423), investment (ρ = 0.387), and Firm performance (ρ = 0.336). These findings suggest that firms characterised by higher levels of OQ are more likely to obtain external finance, undertake greater levels of investment, and achieve superior economic performance. Particularly noteworthy is the relationship between OQ and access to external finance, which is stronger than the corresponding correlation between firm size and access to external finance (ρ = 0.331). Although these correlations do not permit causal inference, they provide preliminary empirical support for the theoretical argument that organisational information constitutes an important input in lenders' credit assessment of SMEs, beyond traditional firm-size characteristics. Moreover, the correlations among the explanatory variables remain sufficiently moderate to rule out serious multicollinearity concerns. This conclusion is further supported by the variance inflation factor (VIF) values, all of which are below the conservative threshold of 3 and well below the conventional cut-off value of 5 commonly adopted in the econometric literature. Consequently, no evidence emerges of multicollinearity issues that could undermine the stability of the subsequent structural model estimates. While these findings provide encouraging preliminary support for the proposed theoretical framework, they are not sufficient to test the central proposition of the present study, namely that the various organisational signals represent observable manifestations of a single latent construct, OQ. It is therefore necessary to validate the measurement model through EFA and CFA to assess its reliability, convergent validity, and discriminant validity before estimating the structural model using SEM.

Table 6. Correlation matrix

Variable	OQ	Access to External Finance	Investment	Firm Performance	Firm Size	Firm Age	VIF
OQ	1.000	—	—	—	—	—	—
Observed access to external finance	0.423***	1.000	—	—	—	—	—
Observed investment intensity	0.387***	0.481***	1.000	—	—	—	—
Firm performance	0.336***	0.274***	0.359***	1.000	—	—	—
Firm size	0.312***	0.331***	0.286***	0.191***	1.000	—	1.94
Firm age	0.147***	0.182***	0.116***	0.093***	0.204***	1.000	1.53

Note: OQ = Organisational Quality; VIF = variance inflation factor. *** $p < 0.01$; "—" indicates that there is no data.

Table 7 reports the results of the validation procedure for the OQ construct. Consistent with the theoretical framework developed in Section 2, the construct is conceptualised as a reflective latent variable that is not directly observable but is manifested through a range of indicators capturing organisational structure, managerial capabilities, innovation, digitalisation, human capital quality, relational networks, and financial transparency. The validation procedure was conducted in two stages. First, an EFA was performed to examine whether the selected indicators shared a common underlying factor structure. Subsequently, a CFA was estimated to evaluate the reliability of the measurement model and the overall quality of the latent construct.

Table 7. Summary validation of the Organisational Quality construct

Indicator	Value
Kaiser-Meyer-Olkin (KMO)	0.872
Bartlett's test χ^2	4,986.42
Bartlett *p*-value	<0.001
Total variance explained	67.3%
Cronbach's alpha	0.892
Composite reliability (CR)	0.918
Average variance extracted (AVE)	0.620
Minimum factor loading	0.681
Maximum factor loading	0.861
Comparative fit index (CFI)	0.968
Tucker–Lewis index (TLI)	0.962
Root mean square error of approximation (RMSEA)	0.036
Standardised root mean square residual (SRMR)	0.034

The evidence reported in Table 7 confirms the robustness of the measurement model. The Kaiser-Meyer-Olkin (KMO) statistic (0.872) indicates that the data are highly suitable for factor analysis, while Bartlett's test of sphericity is highly significant ($p < 0.001$), allowing the null hypothesis of no correlation among the observed indicators to be rejected. EFA identifies a single dominant factor explaining more than 67% of the total variance, suggesting that the selected indicators share a common underlying structure consistent with the theoretical concept of OQ. CFA provides further support for this interpretation. All factor loadings exceed the threshold of 0.60 commonly recommended in the literature, indicating that each indicator contributes meaningfully to the measurement of the latent construct. Similarly, Cronbach's alpha (0.892) and composite reliability (CR) (0.918) demonstrate a high level of internal consistency, while the average variance extracted (AVE) of 0.620 comfortably exceeds the recommended threshold of 0.51, thereby confirming the convergent validity of the construct. The model fit indices likewise indicate an excellent fit between the measurement model and the observed data. The CFI (0.968) and the TLI (0.962) both exceed the recommended benchmark of 0.95, whereas the RMSEA (0.036) and the SRMR (0.034) remain well below the conventional cut-off values for a well-fitting model. Taken together, these findings demonstrate that the proposed measurement model provides an accurate representation of the observed data and that the OQ construct exhibits satisfactory psychometric properties. From a theoretical perspective, these findings constitute one of the principal contributions of the present study. Previous research has generally examined corporate governance, innovation, human capital, digitalisation, financial transparency, and relational networks as independent determinants of firms' access to external finance or economic performance. By contrast, the present evidence suggests that these organisational characteristics share a common underlying organisational dimension that can be empirically captured through the latent construct of OQ. This implies that their economic effects depend not only on the contribution of each characteristic but also on the overall OQ jointly reflected by these dimensions. This interpretation extends the traditional perspective of signalling theory. Rather than viewing each organisational characteristic as an independent signal, the results indicate that these characteristics collectively reflect a broader organisational capability perceived by external financiers. OQ therefore emerges as the underlying mechanism through which multiple organisational dimensions jointly reduce information asymmetries and facilitate firms' access to external financial resources.

Table 8 reports the results of the CFA for the three reflective latent constructs included in the proposed SEM framework: OQ, access to external finance, and investment. All standardised factor loadings exceed the recommended threshold of 0.60 and are statistically significant at the 1% level, providing strong evidence of convergent validity. For OQ, the highest loadings are observed for accounting discipline, quality certification, and standards adoption, suggesting that financial transparency and organisational formalisation represent particularly important manifestations of the underlying organisational construct. Similarly, all indicators associated with access to external finance and investment load strongly to their respective latent constructs, supporting the adequacy of the proposed measurement model.

Table 8. Confirmatory factor analysis of the reflective latent constructs

Construct	Indicator	Standardised Loading
Organisational Quality	MGR_EDU	0.742
	MGR_PROF	0.791
	MGR_EXP	0.684
	MGR_OCC	0.706
	QUALITY_CERT	0.842
	STANDARDS	0.813
	PC_INTENSITY	0.754
	E_SALES	0.719
	E_PURCHASE	0.681
	EMP_EDU	0.775
	EMP_TRAIN	0.731
	EMP_EXP	0.698
	NETWORK	0.782
	ACCOUNT_DISCIPLINE	0.861
Access to external finance	BANK_LOAN	0.824
	FORMAL_FINANCE	0.791
	BANK_RELATIONSHIP	0.736
	FIN_CONSTRAINT	0.701
Investment	MACHINERY	0.845
	PLANT	0.817
	EQUIPMENT	0.769
	FIXED_ASSETS	0.734

Note: All standardised factor loadings are statistically significant at the 1% level.

Table 9 reports the reliability and convergent validity assessment of the three reflective latent constructs. The results indicate excellent internal consistency, as Cronbach's alpha, McDonald's Omega, and CR all exceed the recommended threshold of 0.70. In addition, the AVE is above the recommended threshold of 0.50 for all constructs, providing further evidence of convergent validity. Taken together, these findings confirm that the latent constructs are measured with a satisfactory degree of reliability and validity.

Table 9. Reliability and convergent validity of the reflective latent constructs

Construct	Cronbach's Alpha	McDonald's Omega	CR	AVE
Organisational Quality	0.89	0.92	0.91	0.62
Access to external finance	0.86	0.87	0.88	0.59
Investment	0.84	0.85	0.87	0.61

Note: CR = composite reliability; AVE = average variance extracted; Kaiser–Meyer–Olkin = 0.87; Bartlett's test of sphericity: χ^2 = 4,986.42, $p < 0.001$. The Kaiser–Meyer–Olkin statistic and Bartlett's test refer to the organisational indicators used to construct Organisational Quality.

Both the Fornell–Larcker criterion (Table 10) and Heterotrait–Monotrait (HTMT) ratios (Table 11) provide evidence of discriminant validity. The square root of AVE exceeds the corresponding inter-construct correlations, and all HTMT values remain well below the conservative threshold of 0.85.

Off-diagonal elements in Table 10 report the correlations among the latent constructs estimated within the CFA measurement model. These values may differ slightly from the descriptive Pearson correlations reported in Table 6 because they are estimated at the latent-variable level and therefore account for measurement error. Diagonal elements report the square root of the AVE.

Table 10. Discriminant validity assessment (Fornell–Larcker criterion)

Construct	OQ	Access to External Finance	Investment
OQ	0.787	—	—
Access to external finance	0.460	0.768	—
Investment	0.390	0.490	0.781

Note: OQ = Organisational Quality; AVE = average variance extracted; Diagonal values = √AVE; "—" indicates that there is no data.

Table 11. Heterotrait–Monotrait (HTMT) ratios

Construct	HTMT
Organisational Quality (OQ)–Access to external finance	0.62
OQ–Investment	0.55
Access to external finance–Investment	0.68

Having validated the measurement model, the structural model was subsequently estimated using SEM. This methodology makes it possible to analyse the relationships among latent constructs simultaneously while explicitly accounting for measurement error and estimating both direct and indirect effects. Consequently, the proposed theoretical framework can be tested empirically by examining whether OQ contributes to value creation through improved access to external finance and the subsequent increase in productive investment.

The results of the structural model are presented in Table 12.

Overall, the model fit indices indicate that the proposed model provides an excellent representation of the observed data structure. All fit statistics meet the commonly accepted thresholds reported in the methodological literature, confirming the overall adequacy of the structural model. The most important finding concerns the positive relationship between OQ and access to external finance. The estimated standardised coefficient ($\beta = 0.427$; $p < 0.001$) is the largest in the entire model and provides strong support for the hypothesis that higher OQ reduces information asymmetries and enhances firms' ability to obtain external financial resources. From an economic perspective, a one-standard-deviation increase in OQ is associated with an increase of approximately 0.43 standard deviation in access to external finance, even after controlling for firm size, firm age, export orientation, industry affiliation, and geographical location. From a practical perspective, this finding suggests that investments in organisational capabilities—such as improving governance structures, strengthening managerial and workforce skills, adopting digital technologies, enhancing financial transparency, and developing stronger business relationships—may substantially increase SMEs' credibility in the eyes of external financiers. Consequently, organisational improvements should be viewed not only as internal efficiency-enhancing practices but also as strategic investments that facilitate firms' access to external financial resources. Rather than identifying a new determinant of access to external finance, the analysis proposes an alternative conceptualisation of firms' organisational characteristics. Whereas the existing literature typically treats governance, innovation, digitalisation, human capital, and relational networks as distinct determinants, the evidence suggests that these dimensions share a common underlying organisational component, captured by OQ. OQ therefore emerges as a latent organisational dimension reflected by multiple observable organisational characteristics. Firms characterised by higher levels of

OQ provide a richer and more coherent set of organisational signals, thereby reducing information asymmetries and facilitating lenders' assessment of firms' creditworthiness.

Table 12. Structural equation modelling results

Relationship	Standardised β	S.E.	z	p-Value
OQ → Access to external finance	0.427	0.051	8.37	<0.001
Access to external finance → Investment	0.351	0.046	7.63	<0.001
Investment → Firm performance	0.281	0.048	5.85	<0.001
OQ → Investment	0.158	0.052	3.04	0.002
OQ → Firm performance	0.070	0.033	2.15	0.031
Firm size → Access to external finance	0.213	0.034	6.29	<0.001
Firm age → Access to external finance	0.097	0.028	3.46	0.001
Export Orientation → Firm performance	0.142	0.031	4.58	<0.001
Model Fit Indices				
$\chi^2 = 624.83$				
$df = 286$				
$\chi^2/df = 2.18$				
CFI = 0.964				
TLI = 0.959				
IFI = 0.964				
RMSEA = 0.037				
SRMR = 0.035				

Note: S.E. = standard error; z = z-statistic; OQ = Organisational Quality.

The model also reveals a positive and statistically significant relationship between access to external finance and investment ($\beta = 0.351$; $p < 0.001$). This finding confirms that the availability of external capital constitutes a fundamental prerequisite for supporting SMEs' investment activities. In practical terms, easier access to external finance enables firms to undertake investments in productive assets that might otherwise be postponed or abandoned because of financial constraints. This result highlights the importance of reducing financing frictions to support firms' long-term growth and competitiveness. However, the present study suggests that this relationship should be interpreted within a broader causal mechanism, whereby the availability of financial resources depends, at least in part, on lenders' assessment of the firm's OQ. Consistent with theoretical expectations, Investment exerts a positive and statistically significant effect on Performance ($\beta = 0.281$; $p < 0.001$). From a practical perspective, this result indicates that the benefits of improved access to external finance materialise only when the additional financial resources are effectively transformed into productive investments. Simply obtaining external finance is therefore insufficient to enhance firm performance; value creation depends on firms' ability to allocate these resources to investments that strengthen productive capacity and support long-term growth. This evidence indicates that the financial resources obtained are effectively transformed into productive investments capable of improving firms' economic performance. Taken together, these findings provide initial empirical support for the proposed causal pathway, whereby OQ promotes value creation by improving access to external finance, which subsequently stimulates investment. The model further identifies a positive direct effect of OQ on Investment ($\beta = 0.158$; $p = 0.002$). This suggests that firms characterised by higher OQ tend to invest more even independently of their access to external finance. One possible explanation is that such firms possess superior managerial capabilities in strategic planning, opportunity identification, and the efficient allocation of available resources. By contrast, the direct effect of OQ on Performance is positive but relatively modest ($\beta = 0.070$; $p = 0.031$). This result is theoretically important because it suggests that the economic value of OQ does not arise primarily through an immediate impact on firm performance, but rather through a mediated process. In other words, OQ first enhances the firm's ability to obtain external finance, subsequently facilitates investment activity, and only through this sequential mechanism contributes to improved economic performance. Overall, the empirical evidence is fully consistent with the proposed theoretical framework. The findings suggest that OQ represents a latent organisational dimension capable of explaining how firms' organisational characteristics influence access to external finance, investment decisions, and, indirectly, value creation. The relatively modest direct effect of OQ on performance, compared with its substantially stronger indirect effects, further indicates that the economic value of OQ is realised primarily through improvements in firms' financing and investment processes. This reinforces the view that OQ constitutes a strategic organisational resource whose contribution to firm performance operates predominantly through its influence on capital allocation mechanisms. From a practical perspective, these findings suggest that investments in organisational capabilities—including governance quality, human capital, innovation, digitalisation, financial transparency, and relational networks—should not be viewed solely as internal managerial improvements. Rather, they represent strategic investments that enhance firms' credibility in the eyes of external financiers, facilitate access to financial resources, and ultimately support long-term value creation. Likewise, the results indicate that financial institutions may benefit from incorporating organisational signals alongside traditional financial information when assessing SMEs' creditworthiness.

To formally test the proposed theoretical mechanism, the direct and indirect effects implied by the structural model were estimated using a bias-corrected bootstrap procedure with 5,000 resamples. This approach, which is widely adopted in the mediation literature, provides robust confidence intervals for indirect effects without requiring the assumption of normality, thereby enhancing the reliability of the statistical inferences. The results of the mediation analysis are reported in Table 13.

Table 13. Direct and indirect effects

Effect	Coefficient	Bootstrap S.E.	95% CI	*p*-Value
Direct effect OQ → Performance	0.070	0.028	0.016–0.124	0.012
Indirect effect OQ → Access to external finance → Investment	0.150	0.021	0.109–0.191	<0.001
Indirect effect OQ → Investment → Performance	0.050	0.012	0.026–0.074	<0.001
Sequential mediation OQ → Access to external finance → Investment → Performance	0.050	0.011	0.028–0.072	<0.001

Note: OQ = Organisational Quality; S.E. = standard error; CI = confidence interval.

The findings indicate that the effect of OQ on firm performance is transmitted not only through a direct effect but also through indirect mechanisms involving firms' financing and investment decisions. The bootstrap mediation analysis confirms that the direct effect of OQ on firm performance remains positive and statistically significant ($\beta = 0.070$; bootstrap $p = 0.012$; 95% CI = 0.016–0.124), consistently with the significant direct effect identified in the structural model. The indirect effects are likewise positive and highly significant, suggesting that OQ creates value primarily by strengthening firms' ability to obtain external financial resources and subsequently transform those resources into productive investment. The mediation analysis further indicates that improved access to external finance represents an important transmission mechanism through which OQ contributes to firm performance by creating the financial conditions necessary to support firms' investment activity. More specifically, the indirect effect of OQ on Investment through access to external finance is positive and statistically significant ($\beta = 0.150$; 95% CI = 0.109–0.191; $p < 0.001$), indicating that improved access to external finance represents a significant channel through which OQ supports firms' investment activity.

Particularly noteworthy is the sequential mediation pathway OQ → Access to external finance → Investment → Firm performance, which yields a statistically significant standardised coefficient ($\beta = 0.050$; $p < 0.001$). This finding provides strong empirical support for the theoretical framework developed in the preceding sections and confirms that OQ contributes to firm performance through a sequential process. Specifically, higher OQ reduces the information asymmetries perceived by external financiers, facilitates access to external finance, promotes productive investment, and ultimately enhances firm performance. Overall, the mediation analysis indicates that OQ contributes to firm performance primarily through firms' financing and investment processes rather than solely through a direct effect on performance. Access to external finance therefore represents the first transmission mechanism linking OQ to firm performance, while productive investment constitutes the channel through which the financial resources obtained are translated into economic value creation.

These findings have important theoretical implications. Had the effect of OQ been predominantly direct, OQ could simply have been interpreted as another determinant of firm performance. Instead, the empirical evidence suggests a broader interpretation, demonstrating that the principal economic value of OQ lies in its capacity to improve the functioning of capital allocation mechanisms and to facilitate the transformation of financial resources into productive investment. From a corporate finance perspective, these findings extend the traditional information asymmetry framework. The classical literature has generally explained firms' access to external finance primarily through observable financial characteristics, such as accounting information, collateral, firm size, and profitability. The present evidence suggests that OQ constitutes an additional strategic capability that enhances firms' credibility in the eyes of external financiers, facilitates access to external capital, and ultimately supports productive investment. Rather than assigning separate effects to corporate governance, human capital, innovation, digitalisation, financial transparency, and relational networks, the proposed framework demonstrates that these organisational characteristics reflect a common latent organisational dimension. OQ therefore emerges as the underlying mechanism through which these complementary organisational attributes jointly improve firms' access to external finance, foster productive investment, and ultimately contribute to value creation.

Table 14 reports the results of the heterogeneity analysis. The informational value of OQ increases as informational opacity rises. The effect is strongest among younger firms, smaller firms, and firms characterised by lower collateral availability. Consistent with H2, the effect of OQ on access to external finance is significantly stronger among firms operating under greater information asymmetry. In particular, the estimated coefficient is substantially larger for young firms (β = 0.550) than for mature firms (β = 0.340; Wald test, p = 0.006). A similar pattern emerges when firms are classified according to size and collateral availability. Specifically, the effect of OQ is stronger for small firms (β = 0.490) than for large firms (β = 0.310; Wald test, p = 0.015), and reaches its highest value among firms with low collateral (β = 0.570), compared with firms characterised by high collateral (β = 0.290; Wald test, p = 0.002). These findings provide direct empirical support for H2 and reinforce the theoretical interpretation proposed in this study. Rather than representing simply another determinant of access to external finance, OQ operates as an informational mechanism whose economic value increases as information asymmetry becomes more severe.

Table 14. Heterogeneity analysis: the conditional effect of information asymmetry

Subsample	Organisational Quality → Access to External Finance (β)	$\Delta\beta$	Wald Test (p-Value)
Young firms	0.550***	—	—
Mature firms	0.340***	0.210	p = 0.006
Small firms	0.490***	—	—
Large firms	0.310***	0.180	p = 0.015
Low collateral	0.570***	—	—
High collateral	0.290***	0.280	p = 0.002

Note: Number of observations = 6,051; *** p < 0.01. Reported coefficients are standardised structural equation modelling estimates; $\Delta\beta$ denotes the difference in structural coefficients between the corresponding subsamples. Wald tests assess the statistical significance of these differences; "—" indicates that there is no data.

The reflective specification consistently outperforms the formative alternative (Table 15). The superior fit indices support the theoretical assumption that governance, human capital, digitalisation, innovation, transparency, and relational networks operate as manifestations of an underlying organisational capability rather than as independent formative components.

Table 15. Reflective versus formative model comparison

Model Specification	Comparative Fit Index	Tucker–Lewis Index	Root Mean Square Error of Approximation	Standardised Root Mean Square Residual
Reflective Organisational Quality (OQ)	0.967	0.961	0.039	0.034
Formative OQ	0.905	0.892	0.076	0.072

Finally, to assess the robustness of the findings, additional analyses were conducted using alternative measurement approaches, alternative model specifications, and panel-data models. The results, reported in Table 16, confirm the stability of the estimated relationships and indicate that the conclusions are not driven by specific methodological choices.

Table 16. Robustness analyses

Alternative Specification	OQ → Access to External Finance	Access to External Finance → Investment	Investment → Performance	OQ → Performance	N	Adjusted R^2
Baseline structural equation modelling model	0.427***	0.351***	0.281***	0.070**	6,051	0.412
OQ construct based on PCA	0.401***	0.346***	0.279***	0.081**	6,051	0.407
Logistic regression (Access to external finance)	0.438***	—	—	—	6,051	0.391
Random-effects model	0.416***	0.342***	0.271***	0.079**	6,051	0.401
Fixed-effects model	0.389***	0.326***	0.259***	0.071*	6,051	0.419
Lagged OQ	0.364***	0.317***	0.246***	0.068*	4,034	0.382

Note: OQ = Organisational Quality; N = number of observations; PCA = principal component analysis; * p < 0.10; ** p < 0.05; *** p < 0.01; "—" indicates that there is no data.

To further assess the robustness of the empirical findings, a series of additional analyses was undertaken. The first robustness check concerned the construction of the OQ construct. Although the main analysis relies on a CFA measurement model, the construct was also reconstructed using principal component analysis (PCA), a widely adopted technique for synthesising multidimensional information. The results are highly consistent with those obtained from the baseline model, both in terms of the sign and magnitude of the estimated coefficients, indicating that the study's conclusions are not driven by the specific measurement approach adopted. This evidence strengthens the interpretation of OQ as an economically meaningful latent construct rather than a statistical artefact generated by the estimation procedure. A second robustness test examined the sensitivity of the findings to the treatment of the access-to-finance variable. Given that access to external finance is measured as a binary outcome, the baseline SEM specification was compared with a Logit model in which the probability of obtaining external finance constituted the dependent variable. The estimated coefficient associated with OQ remained positive, substantial, and highly statistically significant ($\beta = 0.438$; $p < 0.001$), confirming that the relationship identified is not contingent upon the use of the SEM framework. To account for potential unobserved firm heterogeneity, panel-data models with both random effects and fixed effects were subsequently estimated. Although the introduction of firm fixed effects led to a modest reduction in the magnitude of the estimated coefficients, all principal relationships retained the same direction and remained statistically significant. These findings suggest that the effect of OQ cannot be attributed solely to time-invariant unobserved firm characteristics but instead represents an independent determinant of financing decisions, investment behaviour, and subsequent firm performance. An additional robustness check addressed the potential issue of reverse causality. It is plausible that firms exhibiting superior performance may subsequently invest more heavily in improving their OQ. To mitigate this concern, a lagged measure of OQ was employed, assuming that organisational characteristics observed in one period influence financing and investment decisions in the subsequent period. Even under this specification, the coefficient associated with OQ remained positive and statistically significant, suggesting that the estimated relationships are unlikely to be driven exclusively by simultaneity. Overall, the robustness analyses provide strong evidence of the stability of the empirical results. The relationships between OQ, access to external finance, investment, and firm performance remain remarkably consistent across alternative measurement approaches and econometric specifications. Importantly, although the magnitude of the estimated coefficients varies only modestly across the different robustness tests, their direction, statistical significance, and economic interpretation remain unchanged. This consistency indicates that the positive role of OQ in improving firms' access to external finance, promoting productive investment, and ultimately enhancing firm performance is not driven by a particular empirical specification, thereby reinforcing the robustness of the proposed theoretical framework. These findings reinforce the credibility of the proposed theoretical framework and suggest that the role of OQ in facilitating access to external financial resources and promoting subsequent value creation reflects a structural economic phenomenon rather than the outcome of a particular statistical specification.

To further strengthen the reliability of the empirical evidence, additional diagnostic tests were conducted to assess the potential presence of endogeneity, measurement bias, and influential observations. Although no observational model can eliminate these concerns, the combined application of multiple diagnostic procedures provides a rigorous assessment of whether such issues are likely to compromise the validity of the estimated relationships. The results of these additional robustness tests are reported in Table 17.

Table 17. Additional robustness and diagnostic tests

Test	Result
Maximum variance inflation factor	2.11
Harman's single-factor test	31.8%
Fornell–Larcker criterion	Satisfied
Maximum heterotrait–monotrait ratio	0.68
Bias-corrected bootstrap (5,000 resamples)	Confirmed
Maximum Cook's distance	0.34
Maximum leverage value	0.12

The additional diagnostic analyses further support the robustness of the empirical model. First, the VIF values are well below the thresholds commonly adopted in the literature, indicating that multicollinearity does not represent a concern. About potential common method bias arising from the use of a single data source, Harman's Single-Factor Test shows that the first factor explains only 31.8% of the total variance, substantially below the conventional 50% threshold. This finding suggests that the estimated relationships are unlikely to be driven by systematic measurement bias. To further assess the potential influence of common method bias, a Common Latent Factor (CLF) procedure was estimated by including an unmeasured latent method factor loading on all reflective indicators. The inclusion of the CLF did not materially alter the standardised factor loadings (maximum $\Delta < 0.20$), indicating that common method bias is unlikely to materially affect the measurement model or the estimated structural relationships. This finding is consistent with the results of Harman's single-factor test, which likewise suggests that common method variance does not represent a serious concern in the present study.

The adequacy of the measurement model is further supported by the assessment of discriminant validity. Both the Fornell–Larcker criterion and the HTMT ratio satisfy the recommended cut-off values, confirming that OQ constitutes an empirically distinct construct from the other latent variables included in the structural model. The analysis of influential observations also provides reassuring evidence. The maximum values of Cook's Distance and Leverage remain well below the conventional critical thresholds, indicating that the estimated coefficients are not disproportionately influenced by a limited number of extreme observations but instead reflect the overall behaviour of the sample. Finally, the bias-corrected bootstrap procedure based on 5,000 resamples confirms the robustness of the estimated mediation effects, providing additional support for the validity of the proposed causal mechanism. Taken together, these diagnostic tests provide further evidence of the reliability of the empirical findings. The relationships linking OQ, access to external finance, investment, and firm performance remain stable across alternative econometric specifications, measurement approaches, and diagnostic procedures. Although, as in any observational study, the possibility of residual endogeneity cannot be entirely ruled out, the evidence indicates that any remaining bias is unlikely to alter the substantive conclusions of the analysis. Overall, the robustness checks reinforce the proposed theoretical interpretation, suggesting that OQ represents an economically meaningful latent construct through which firms mitigate information asymmetries faced by external financiers, improve access to external financial resources, and indirectly enhance value creation.

Although the longitudinal structure of the survey and the SEM framework mitigate several sources of estimation bias, additional analyses were conducted to assess potential endogeneity concerns. First, a lagged version of OQ was employed. Second, an instrumental-variable specification was estimated using lagged provincial averages of organisational indicators as instruments for firm-level OQ. The first-stage F-statistic exceeded the conventional threshold of 10 ($F = 24.6$), confirming instrument relevance. The second-stage estimates remain highly consistent with the baseline SEM results, with the coefficient linking OQ to access to external finance remaining positive and statistically significant ($\beta = 0.41$; $p < 0.001$). These findings suggest that reverse causality and omitted-variable bias are unlikely to drive the main results.

The change in CFI remains well below the recommended threshold of 0.01, confirming measurement invariance across survey waves (Table 18).

Table 18. Measurement invariance across survey waves

Model	Comparative Fit Index (CFI)	ΔCFI
Configural	0.967	—
Metric	0.964	0.003
Scalar	0.960	0.004

Note: "—" indicates that there is no data.

Table 19 summarises the empirical assessment of the research hypotheses. Overall, the empirical evidence provides consistent support for the proposed theoretical framework. All six hypotheses receive empirical support, reinforcing the interpretation of OQ as the latent organisational construct through which firms reduce information asymmetries, improve access to external finance, promote productive investment, and ultimately enhance firm performance.

Table 19. Summary of hypothesis testing

Hypothesis	Expected Relationship	Empirical Evidence	Decision
H1	OQ → Access to external finance (+)	$\beta = 0.427$***	Supported
H2	The positive effect of OQ on Access to external finance is stronger under higher information asymmetry	Multigroup SEM (Young, small, low collateral): $\Delta\beta$ significant	Supported
H3	Access to external finance → Investment (+)	$\beta = 0.351$***	Supported
H4	Investment → Firm performance (+)	$\beta = 0.281$***	Supported
H5	Access to external finance mediates the relationship between OQ and investment	Indirect effect significant (bootstrap)	Supported
H6	OQ affects firm performance sequentially through access to external finance and investment	Sequential indirect effect significant (bootstrap)	Supported

Note: OQ = Organisational Quality; SEM = structural equation modelling; *** $p < 0.01$; Mediation effects are assessed using bias-corrected bootstrap confidence intervals based on 5,000 resamples.

5. Discussion and Conclusions

This study developed and empirically validated a novel theoretical framework explaining how OQ influences SMEs' access to external finance. Unlike the existing literature, which typically examines corporate governance, innovation, human capital, digitalisation, relational networks, and information transparency as separate determinants of financing decisions or firm performance, the proposed framework conceptualised these organisational characteristics as observable manifestations of a single underlying latent construct. In this perspective, OQ represents the overall organisational capability of the firm, providing an integrated explanation of how multiple organisational resources jointly influence financing outcomes. The empirical evidence obtained through SEM strongly supports this interpretation. OQ emerges as a significant determinant of access to external finance, while its effect on firm performance operates predominantly through a sequential mediation process. Firms characterised by higher OQ experience easier access to external finance, which enables greater investment activity and ultimately translates into superior economic performance. The decomposition of direct and indirect effects demonstrated that most of the overall impact of OQ on firm performance is transmitted through financing and investment decisions, suggesting that the principal economic value of OQ lies in its ability to improve capital allocation rather than generating immediate performance gains. These findings can extend both corporate finance and signalling theory. Previous research has generally viewed governance quality, innovation, managerial capabilities, digitalisation, transparency, and relational capital as independent informational signals assessed separately by external financiers. The present study suggests a different interpretation. Rather than representing isolated determinants, these organisational attributes appear to reflect a broader latent organisational dimension that financial institutions evaluate collectively when assessing firm quality under conditions of information asymmetry. OQ therefore provides a unifying theoretical construct that explains how multiple organisational signals combine to shape financing decisions.

The principal theoretical contribution of this study lies precisely in this shift in perspective. Rather than adding another explanatory variable to an already extensive list of determinants of access to external finance, the proposed framework offers a higher-level conceptualisation capable of integrating several previously disconnected research streams. Corporate governance, human capital, innovation, digital transformation, information quality, and signalling are no longer interpreted as competing or independent explanations, but as complementary dimensions of a common organisational capability. This integrated perspective contributes to the literature by providing a more parsimonious explanation of how firms reduce information asymmetries and improve their financial credibility.

The findings also have important managerial implications. For SME managers, investments aimed at strengthening governance structures, developing managerial competencies, enhancing digital capabilities, improving workforce skills, expanding relational networks, and increasing information transparency should not be viewed as isolated organisational initiatives. Instead, these investments collectively enhance the firm’s OQ, thereby improving its ability to obtain external finance and sustain long-term growth. The empirical evidence suggests that organisational improvements should therefore be considered strategic investments capable of reducing information asymmetries and strengthening firms’ financial credibility, rather than simply mechanisms for improving internal efficiency. For banks and other financial institutions, the results suggest that incorporating organisational indicators into credit assessment models may complement traditional financial information and provide a more comprehensive evaluation of firms characterised by limited accounting disclosure or short credit histories. Considering organisational signals alongside conventional financial indicators may improve lenders’ ability to identify firms with stronger long-term growth potential and allocate financial resources more efficiently under conditions of information asymmetry. From a public policy perspective, the evidence indicates that initiatives designed to improve SMEs’ access to external finance should extend beyond expanding credit supply to include programmes that strengthen firms’ organisational capabilities, as these capabilities appear to generate indirect economic benefits through improved financing opportunities.

Several limitations provide opportunities for future research. First, the empirical analysis is based on Vietnamese manufacturing SMEs operating within an institutional environment characterised by relatively high information asymmetry and a strong dependence on bank financing. Consequently, the generalisability of the findings to firms operating in other countries, industries, and institutional environments should be interpreted with caution. Although the theoretical framework proposed in this study is expected to be applicable beyond the Vietnamese manufacturing context, its empirical validity should be further examined using data from SMEs operating in different sectors, financial systems, and institutional settings. Cross-country and cross-industry comparative studies would help determine whether the relationships between OQ, access to external finance, investment, and firm performance remain stable under different economic and institutional conditions. Second, information asymmetry is inherently unobservable and is therefore proxied using firm age, firm size, and collateral availability, consistent with established practices in the corporate finance literature. Although these proxies capture important dimensions of informational opacity, they cannot fully represent the complexity of firms’ information environments. Future research could employ more direct measures of information asymmetry or exploit exogenous changes in information availability to provide additional evidence on the conditional financing value of OQ. Finally, although the extensive robustness analyses substantially reduce concerns regarding model specification and endogeneity, the observational nature of the data does not permit definitive causal inference. Overall, this study demonstrates that OQ represents a meaningful latent organisational construct through which firms reduce information asymmetries, improve access to external finance, and ultimately enhance value creation. Rather than identifying another independent determinant of financing decisions, the study introduces a new theoretical perspective in which organisational characteristics traditionally examined in isolation are interpreted as observable manifestations of a common latent organisational dimension. By integrating organisational resources, information asymmetry, financing decisions, investment, and firm performance within a unified theoretical framework, the study advances the corporate finance literature and provides a foundation for future research on the organisational determinants of firms' financial development and long-term value creation.

Data Availability

The data used in this study are derived from the Vietnam Small and Medium Enterprise Survey and are available from the United Nations University World Institute for Development Economics Research (UNU-WIDER), subject to the applicable data access conditions.

Conflicts of Interest

The author declares no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

The author used generative artificial intelligence tools to assist with language editing, stylistic refinement, and manuscript preparation. The author takes full responsibility for the content of the manuscript, including its accuracy, originality, interpretation, and conclusions.

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